Vanguard International Semiconductor Co

Robust 8" demand and pricing with solid 12" fab ramp

Reiterate Rating: BUY | PO: 250.00 TWD | Price: 132.00 TWD

3Q sales/margin supported by PMIC strength

Vanguard's 3Q26 sales rose 10.2% QoQ on the back of increased wafer shipments (up 5% QoQ), particularly for server and smartphone PMIC, and rising wafer prices (up 3% QoQ). As higher utilization of around 90% and a more favorable pricing and product mix more than offset rising manufacturing costs, the 3Q26 gross margin improved by 130bps QoQ, to 33.6%, bringing 3Q26 EPS to NT$1.62 (vs. BofAe/Street of NT$1.67/1.58).

4Q capacity bottleneck offset by ASP uplift

The company guided to 4Q26 sales growth of 5% QoQ at the midpoint, driven by a 1% QoQ increase in wafer shipments, with strong demand capped to some degree by tight capacity, and 4% higher ASPs QoQ, the stronger pricing reflecting rising costs. Assuming further utilization improvement above 90% amid this pricing strength, Vanguard expects the 4Q26 gross margin to expand to 35% at the midpoint, in line with BofAe.

Solid 8" growth and smooth 12" ramp drive '27-28 growth

We believe Vanguard's 8-inch utilization is on track to improve to the mid- to high 90s (%) in 1H27 and remain fully loaded from 2H27 on solid demand related to power IC for computing applications, while 12-inch VSMC should enter mass production from 1Q27, with utilization staying above 90% as capacity gradually expands to 44k WPM by 4Q28. Given the solid commitment from TSMC (interposer demand), the 12-inch margin profile should fare better than previously expected. This, coupled with an 8-inch price hike amid tight supply should drive GM expansion to 36% in 2027 (vs. 32.8% in 2026).

Decent EPS CAGR of 72% supports high valuation

Our EPS forecasts are largely unchanged at NT$6.17 in 2026, NT$10.33 in 2027, and NT$18.15 in 2028. We reiterate our Buy rating and PO of NT$250, still based on 17x 2H27-1H28e EPS (vs. the stock's historical 10-32x range) as a benign 8-inch supply-demand landscape and smooth 12-inch ramp-up support a 45% sales and 72% earnings CAGR over 2026-28e.

Estimates (Dec) (NT$)	2024A	2025A	2026E	2027E	2028E
Net Income (Adjusted - mn)	7,046	7,908	11,917	19,971	35,085
EPS	4.21	4.30	6.17	10.33	18.15
EPS Change (YoY)	-6.4%	2.2%	43.4%	67.6%	75.7%
Dividend / Share	4.50	4.50	4.50	4.50	4.50
Free Cash Flow / Share	5.27	(20.57)	(10.50)	5.48	14.94
Valuation (Dec)					
P/E	31.37x	30.70x	21.41x	12.78x	7.27x
Dividend Yield	3.41%	3.41%	3.41%	3.41%	3.41%
EV / EBITDA*	21.01x	20.17x	15.15x	8.95x	5.68x
Free Cash Flow Yield*	3.56%	-15.25%	-8.19%	4.27%	11.65%

* For full definitions of *iQmethod℠* measures, see page 6.

This research report provides general information only. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or other persons without the express written consent of BofA Securities.
>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.
Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.
BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.
Refer to important disclosures on page 7 to 9. Analyst Certification on page 4. Price

10 November 2026

Equity

Key Changes

(NT$)	Previous	Current
2026E EPS	6.40	6.17
2027E EPS	10.46	10.33
2028E EPS	18.35	18.15
2026E EBITDA (m)	21,705.1	21,607.5
2027E EBITDA (m)	36,140.6	36,579.8
2028E EBITDA (m)	56,733.4	57,663.3

Haas Liu >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3727
haas.liu@bofa.com

Mike Yang >>
Research Analyst
Merrill Lynch (Taiwan)
mike.c.yang@bofa.com

Cathy Hsu >>
Research Analyst
Merrill Lynch (Taiwan)
cathy.hsu3@bofa.com

Stock Data

Price	132.00 TWD
Price Objective	250.00 TWD
Date Established	29-Sep-2026
Investment Opinion	C-1-7
52-Week Range	87.60 TWD-223.00 TWD
Mrkt Val / Shares Out (mn)	7,526 USD / 1,878.5
Market Value (mn)	247,962 TWD
Average Daily Value (mn)	70.05 USD
Free Float	61.4%
BofA Ticker / Exchange	VGILF / TWO
Bloomberg / Reuters	5347 TT / 5347.TWO
ROE (2026E)	19.5%
Net Dbt to Eqty (Dec-2025A)	-40.8%

Abbreviations

DDIC = display driver integrated circuit
IDM = integrated device manufacturer
PMIC = power management integrated circuit
VSMC = VisionPower Semiconductor Manufacturing Co
WPM = wafers per month

iQprofile℠ Vanguard International Semiconductor Co

Key Income Statement Data (Dec)	2024A	2025A	2026E	2027E	2028E
(NT$ Millions)					
Sales	44,055	48,591	59,156	88,161	123,811
Gross Profit	11,933	13,654	19,419	31,742	52,041
Sell General & Admin Expense	(2,740)	(3,691)	(4,312)	(6,262)	(7,936)
Operating Profit	7,111	7,773	12,451	21,786	39,425
Net Interest & Other Income	1,670	1,581	1,438	1,551	1,627
Associates	(29)	(104)	(23)	0	0
Pretax Income	8,752	9,251	13,866	23,337	41,052
Tax (expense) / Benefit	(1,705)	(1,480)	(2,324)	(3,982)	(7,051)
Net Income (Adjusted)	7,046	7,908	11,917	19,971	35,085
Average Fully Diluted Shares Outstanding	1,675	1,839	1,933	1,933	1,933
Key Cash Flow Statement Data					
Net Income	7,046	7,908	11,917	19,971	35,085
Depreciation & Amortization	8,467	8,458	9,157	14,793	18,238
Change in Working Capital	(1,958)	(2,465)	(4,095)	(6,798)	(6,525)
Deferred Taxation Charge	NA	NA	NA	NA	NA
Other Adjustments, Net	11,176	11,539	14,716	(2,166)	(2,711)
Cash Flow from Operations	24,731	25,439	31,696	25,800	44,088
Capital Expenditure	(15,909)	(63,265)	(52,000)	(15,200)	(15,200)
(Acquisition) / Disposal of Investments	3,641	1,118	(868)	0	0
Other Cash Inflow / (Outflow)	570	2,436	2,903	0	0
Cash Flow from Investing	(11,699)	(59,711)	(49,965)	(15,200)	(15,200)
Shares Issue / (Repurchase)	17,593	0	0	0	0
Cost of Dividends Paid	NA	NA	NA	NA	NA
Cash Flow from Financing	17,530	13,460	24,806	(8,699)	(8,699)
Free Cash Flow	8,822	(37,826)	(20,304)	10,600	28,888
Net Debt	(64,793)	(34,122)	(32,490)	(34,778)	(55,054)
Change in Net Debt	(27,857)	28,579	18,127	(1,901)	(20,189)
Key Balance Sheet Data					
Property, Plant & Equipment	44,141	113,869	136,165	138,021	136,378
Other Non-Current Assets	18,353	19,702	19,584	19,598	19,599
Trade Receivables	6,235	6,748	9,153	13,536	18,056
Cash & Equivalents	66,749	41,716	49,481	51,769	72,045
Other Current Assets	13,227	18,433	17,550	20,652	23,549
Total Assets	148,706	200,469	231,933	243,575	269,627
Long-Term Debt	1,710	1,837	16,743	16,743	16,743
Other Non-Current Liabilities	56,953	63,438	82,172	78,096	78,438
Short-Term Debt	245	5,757	248	248	248
Other Current Liabilities	21,130	45,847	44,912	45,413	46,169
Total Liabilities	80,039	116,879	144,075	140,500	141,598
Total Equity	68,667	83,590	87,858	103,075	128,029
Total Equity & Liabilities	148,706	200,469	231,933	243,575	269,627
iQmethod℠ **- Bus Performance***					
Return On Capital Employed	7.0%	5.7%	6.8%	10.1%	16.2%
Return On Equity	12.8%	12.4%	19.5%	30.2%	40.3%
Operating Margin	16.1%	16.0%	21.0%	24.7%	31.8%
EBITDA Margin	35.4%	33.4%	36.5%	41.5%	46.6%
iQmethod℠ **- Quality of Earnings***					
Cash Realization Ratio	3.5x	3.2x	2.7x	1.3x	1.3x
Asset Replacement Ratio	1.9x	7.5x	5.7x	1.0x	0.8x
Tax Rate (Reported)	19.5%	16.0%	16.8%	17.1%	17.2%
Net Debt-to-Equity Ratio	-94.4%	-40.8%	-37.0%	-33.7%	-43.0%
Interest Cover	20.4x	22.9x	NM	NM	NM

Key Metrics

* For full definitions of *iQmethod*℠ measures, see page 6.

Company Sector

Semiconductors

Company Description

Vanguard International Semiconductor Co is a pure 8-inch foundry headquartered in Hsinchu, Taiwan. The company has five fabs, one of which is in Singapore. Collectively, the capacity is now about 280k WPM, and the company focuses on fabricating power semiconductors and driver IC for both IDMs (integrated device manufacturers) and fabless customers.

Investment Rationale

We have a Buy rating on Vanguard as we expect an improvement in the overall 8-inch foundry supply/demand landscape from TSMC's retirement of some legacy capacity and limited new industry capacity additions. The higher utilization and pricing upcycle in 8-inch, coupled with enhanced mix from market share gains in Power IC, are likely to support sales and gross margin expansion.

Stock Data

Price to Book Value	4.3x

Focus tables

Exhibit 1: Vanguard 3Q26-4Q26 and 2026-27 estimates BofAe vs. Street
Our new EPS estimates are NT$6.17 for 2026 and NT$10.33 for 2027

	3Q26E				4Q26E			2026E			2027E	
	Actual	Old	Street	BofAe	Old	Street	BofAe	Old	Street	BofAe	Old	Street
Sales (NT$mn)	$15,702	$15,575	$15,385	$16,678	$17,305	$16,314	$59,156	$59,656	$58,321	$88,161	$88,780	$78,788
Sales QoQ / YoY (%)	**10.2%**	**9.3%**	**8.0%**	**6.2%**	**11.1%**	**6.0%**	**21.7%**	**22.8%**	**20.0%**	**49.0%**	**48.8%**	**35.1%**
Gross profit (NT$mn)	$5,283	$5,260	$5,147	$5,861	$6,050	$5,535	$19,419	$19,584	$18,829	$31,742	$31,837	$25,039
GMs (%)	**33.6%**	**33.8%**	**33.5%**	**35.1%**	**35.0%**	**33.9%**	**32.8%**	**32.8%**	**32.3%**	**36.0%**	**35.9%**	**31.8%**
Operating profit (NT$mn)	$3,470	$3,402	$3,338	$3,994	$4,038	$3,646	$12,451	$12,427	$11,878	$21,786	$21,330	$16,104
OpMs (%)	**22.1%**	**21.8%**	**21.7%**	**23.9%**	**23.3%**	**22.4%**	**21.0%**	**20.8%**	**20.4%**	**24.7%**	**24.0%**	**20.4%**
Pretax profit (NT$mn)	$3,760	$3,753	$3,637	$4,356	$4,373	$3,923	$13,866	$13,875	$13,251	$23,337	$22,754	$17,273
Pretax margins (%)	23.9%	24.1%	23.6%	26.1%	25.3%	24.0%	23.4%	23.3%	22.7%	26.5%	25.6%	21.9%
Tax expense (NT$mn)	-$663	-$636	-$673	-$757	-$753	-$686	-$1,949	-$1,918	-$1,980	-$3,367	-$3,226	-$3,034
Tax rate (%)	17.6%	16.9%	18.5%	17.4%	17.2%	17.5%	14.1%	13.8%	14.9%	14.4%	14.2%	17.6%
Net profit (NT$mn)	$3,097	$3,117	$2,964	$3,599	$3,620	$3,237	$11,917	$11,957	$11,271	$19,971	$19,528	$14,239
Net margins (%)	19.7%	20.0%	19.3%	21.6%	20.9%	19.8%	20.1%	20.0%	19.3%	22.7%	22.0%	18.1%
EPS (NT$)	**$1.62**	**$1.67**	**$1.58**	**$1.86**	**$1.94**	**$1.70**	**$6.17**	**$6.40**	**$6.00**	**$10.33**	**$10.46**	**$7.52**

Source: Bloomberg, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 2: Vanguard's P&L analysis, by 8" and 12" business
We believe Vanguard's 8-inch growth and margin potential will be overshadowed by the ramp-up of 12-inch, which should turn profitable in '28

Vanguard P&L analysis	2026			2027			2028		
	VIS TW	VIS SG	Consolidated	VIS TW	VIS SG	Consolidated	VIS TW	VIS SG	Consolidated
Revenue	**$58,802**	**$354**	**$59,156**	**$74,079**	**$14,083**	**$88,161**	**$88,689**	**$35,122**	**$123,811**
Total COGs	$38,779	$958	$39,737	$44,194	$12,225	$56,419	$45,257	$26,513	$71,770
- Depreciation in COGs	$8,467	$690	$9,157	$8,044	$6,750	$14,793	$7,641	$10,597	$18,238
Gross profit	**$20,023**	**-$604**	**$19,419**	**$29,885**	**$1,857**	**$31,742**	**$43,433**	**$8,608**	**$52,041**
GMs	**34.1%**	**-170.5%**	**32.8%**	**40.3%**	**13.2%**	**36.0%**	**49.0%**	**24.5%**	**42.0%**
Operating Expenses	$6,788	$180	$6,968	$8,330	$1,626	$9,956	$9,032	$3,584	$12,616
Opex / sales %	11.5%	50.8%	11.8%	11.2%	11.5%	11.3%	10.2%	10.2%	10.2%
Operating profit	**$13,235**	**-$784**	**$12,451**	**$21,555**	**$231**	**$21,786**	**$34,400**	**$5,025**	**$39,425**
OpMs	**22.5%**	**-221.3%**	**21.0%**	**29.1%**	**1.6%**	**24.7%**	**38.8%**	**14.3%**	**31.8%**
Net Non-Op gain / loss	$1,416	$0	$1,416	$1,551	$0	$1,551	$1,627	$0	$1,627
Earnings before tax	**$14,650**	**-$784**	**$13,866**	**$23,106**	**$231**	**$23,337**	**$36,028**	**$5,025**	**$41,052**
Effective Tax Expense (Credit)	18.0%	NM	16.8%	18.0%	NM	17.1%	18.0%	11.3%	17.2%
Earnings after tax	**$12,013**	**-$784**	**$11,543**	**$18,947**	**$231**	**$19,356**	**$29,543**	**$4,459**	**$34,001**
Net Earnings attributed to parent	**$12,075**	**-$470**	**$11,917**	**$19,654**	**$139**	**$19,971**	**$32,410**	**$2,675**	**$35,085**
EPS Basic (NT$)	**$6.25**	**-$0.24**	**$6.17**	**$10.17**	**$0.07**	**$10.33**	**$16.77**	**$1.38**	**$18.15**

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Price objective basis & risk

Vanguard International Semiconductor Co (VGILF)
Our PO of NT$250 is based on 17x 2H27-1H28 EPS. This target multiple is slightly below the midpoint of Vanguard's historical range of 10-32x, reflecting both depreciation associated with the Singapore fab and Vanguard's increasing share in the Power IC market amid a utilization and pricing uptrend.

Downside risks:
(1) Higher cost burden if customers demand to move the production base to US and/or Europe.
(2) Slowdown of growth in auto market where the company has 10-20% revenue exposure.

Upside risks:
(1) Stronger-than-expected end market demand for power management IC (PMIC) and/or display driver IC (DDIC).
(2) Expedited market share gain due to the order shift by customers from China-based foundries.

Analyst Certification

I, Haas Liu, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

Special Disclosures

Information on securities which are listed on the exchanges where ML Securities (Taiwan) Limited is not permitted to trade or solicit trades for clients is for informational purposes only and is not a recommendation or a solicitation to trade such securities. ML Securities (Taiwan) Limited will not execute transactions for nor accept orders from clients to trade such securities. Foreign investment in Taiwan securities is regulated and restricted. Currently, foreign investment in Taiwan securities is permitted by investment through: (1) global depository receipts, (2) convertible bonds, (3) mutual funds issued offshore of Taiwan, and (4) a special foreign institutional investors (FINIs) and foreign individual investors (FIDIs) program supervised by the Taiwan SFB whereunder FINIs/FIDIs may apply for investment ID to invest in Taiwan securities by registration with Taiwan Stock Exchange. FINIs will additionally need consent from the foreign exchange authority, ie, the Central Bank of China. In addition to the limitations above, various industry-specific percentage-based limitations on foreign ownership of Taiwan companies (and in some cases prohibitions) may apply. Investments are subject to exchange rate and currency conversion restrictions and risks. Dividends and interest earned by foreign investors' Taiwan securities/instruments are generally subject to a 20% withholding tax. Ordinary shares are not available to ML private client accounts in the U.S.

This report is distributed in Taiwan by Merrill Lynch Securities (Taiwan) Limited, which is regulated by the Taiwan SFB.

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Alchip	ALCPF	3661 TT	Haas Liu
	All Ring	XNQNF	6187 TT	Haas Liu

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	AMEC	XRTOF	688012 CH	Dai Shen
	ASE Technology Holding	XSRIF	3711 TT	Haas Liu
	ASE Technology Holding -ADR	ASX	ASX US	Haas Liu
	ASMPT	ASMVF	522 HK	Simon Woo, CFA
	Aspeed	XLKMF	5274 TT	Mike Yang
	Chroma ATE	CRMJF	2360 TT	Haas Liu
	CXMT	XCXMF	688825 CH	Dai Shen
	Doosan Corp	DOSAF	000150 KS	Simon Woo, CFA
	eMemory	XYLWF	3529 TT	Mike Yang
	GigaDevice	XGXIF	603986 CH	Daley Li, CFA
	Global Unichip Corp.	GBUHF	3443 TT	Haas Liu
	Grand Process Technology Corp	XZOWF	3131 TT	Haas Liu
	Hanmi Semiconductor	HNSIF	042700 KS	Simon Woo, CFA
	Hon Precision	XXPIF	7769 TT	Haas Liu
	Horizon Robotics	HRZRF	9660 HK	Daley Li, CFA
	Iluvatar CoreX	XSIBF	9903 HK	Daley Li, CFA
	Ingenic	XISCF	300223 CH	Dai Shen
	InnoScience Technology	XSCHF	2577 HK	Daley Li, CFA
	JCET Group Co Ltd	XJIEF	600584 CH	Dai Shen
	King Yuan Electronics Corp.	KYUFF	2449 TT	Haas Liu
	Kinik	KIKCF	1560 TT	Haas Liu
	Macronix International	MXICF	2337 TT	Dai Shen
	MediaTek	MDTKF	2454 TT	Haas Liu
	Montage Technology	XRDFF	688008 CH	Daley Li, CFA
	MPI Corporation	XMJCF	6223 TT	Mike Yang
	Nanya Technology	NNYAF	2408 TT	Simon Woo, CFA
	NCE Power	XVFFF	605111 CH	Daley Li, CFA
	OmniVision	XXHQF	603501 CH	Dai Shen
	Phison Electronics	PISNF	8299 TT	Simon Woo, CFA
	Powerchip Semiconductor Manufacturing Co	XCHPF	6770 TT	Haas Liu
	Powertech Technology	PWOGF	6239 TT	Simon Woo, CFA
	Rockchip	XRPXF	603893 CH	Daley Li, CFA
	Samsung C&T	XCDTF	028260 KS	Simon Woo, CFA
	Samsung Elec -G	SSNHZ	SMSN LI	Simon Woo, CFA
	Samsung Electronics	SSNLF	005930 KS	Simon Woo, CFA
	Samsung Electronics Preferred	SSNNF	005935 KS	Simon Woo, CFA
	Silergy Corp.	SLEGF	6415 TT	Mike Yang
	Silicon Motion	SIMO	SIMO US	Simon Woo, CFA
	SK Hynix	HXSCF	000660 KS	Simon Woo, CFA
	SK Hynix ADR	SKHY	SKHY US	Simon Woo, CFA
	SK Square	SKSQF	402340 KS	Simon Woo, CFA
	Taiwan Semiconductor Manufacturing Co.	TSM	TSM US	Haas Liu
	Taiwan Semiconductor Manufacturing Co.	TSMWF	2330 TT	Haas Liu
	Vanguard International Semiconductor Co	VGILF	5347 TT	Haas Liu
	Winbond Electronics	WBEKF	2344 TT	Dai Shen
	WinWay Technology	XWCLF	6515 TT	Mike Yang
	WT Microelectronics	XZOPF	3036 TT	Mike Yang
NEUTRAL				
	ASMedia Technology Inc.	XZSFF	5269 TT	Mike Yang
	Black Sesame Intl Holding	BSIHF	2533 HK	Daley Li, CFA
	Hangzhou Silan Microelectronics	XDFRF	600460 CH	Daley Li, CFA
	Lion Electronics	XDHFF	605358 CH	Dai Shen
	MetaX	XMEZF	688802 CH	Daley Li, CFA
	Novatek	NVKMF	3034 TT	Mike Yang
	Realtek	RLTKF	2379 TT	Haas Liu
	Shenzhen Goodix	XQPLF	603160 CH	Daley Li, CFA
	Soulbrain	XSBOF	357780 KS	Simon Woo, CFA
	VeriSilicon	XMLZF	688521 CH	Daley Li, CFA
UNDERPERFORM				
	Crystal Clear	XPPTF	300655 CH	Dai Shen
	Faraday	FDYTF	3035 TT	Haas Liu
	GlobalWafers	XWLFF	6488 TT	Mike Yang
	Hua Hong Grace	HHUSF	1347 HK	Dai Shen
	LX Semicon	XLXSF	108320 KS	Simon Woo, CFA

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	M31 Technology	XMTZF	6643 TT	Mike Yang
	Maxscend	XMXSF	300782 CH	Daley Li, CFA
	Parade	PRDWF	4966 TT	Mike Yang
	United Microelectronics Corp.	XUMIF	2303 TT	Haas Liu
	United Microelectronics Corp.	UMC	UMC US	Haas Liu
	Win Semiconductors	WSMIF	3105 TT	Haas Liu
	Wonik IPS	XRHQF	240810 KS	Simon Woo, CFA

iQmethod℠ Measures Definitions

Business Performance	Numerator	Denominator
Return On Capital Employed	NOPAT = (EBIT + Interest Income) × (1 – Tax Rate) + Goodwill Amortization	Total Assets – Current Liabilities + ST Debt + Accumulated Goodwill Amortization
Return On Equity	Net Income	Shareholders' Equity
Operating Margin	Operating Profit	Sales
Earnings Growth	Expected 5 Year CAGR From Latest Actual	N/A
Free Cash Flow	Cash Flow From Operations – Total Capex	N/A

Quality of Earnings	Numerator	Denominator
Cash Realization Ratio	Cash Flow From Operations	Net Income
Asset Replacement Ratio	Capex	Depreciation
Tax Rate	Tax Charge	Pre-Tax Income
Net Debt-To-Equity Ratio	Net Debt = Total Debt – Cash & Equivalents	Total Equity
Interest Cover	EBIT	Interest Expense

Valuation Toolkit	Numerator	Denominator
Price / Earnings Ratio	Current Share Price	Diluted Earnings Per Share (Basis As Specified)
Price / Book Value	Current Share Price	Shareholders' Equity / Current Basic Shares
Dividend Yield	Annualised Declared Cash Dividend	Current Share Price
Free Cash Flow Yield	Cash Flow From Operations – Total Capex	Market Cap = Current Share Price × Current Basic Shares
Enterprise Value / Sales	EV = Current Share Price × Current Shares + Minority Equity + Net Debt + Other LT Liabilities	Sales
EV / EBITDA	Enterprise Value	Basic EBIT + Depreciation + Amortization

iQmethod℠ is the set of BofA Global Research standard measures that serve to maintain global consistency under three broad headings: Business Performance, Quality of Earnings, and validations. The key features of iQmethod are: A consistently structured, detailed, and transparent methodology. Guidelines to maximize the effectiveness of the comparative valuation process, and to identify some common pitfalls.

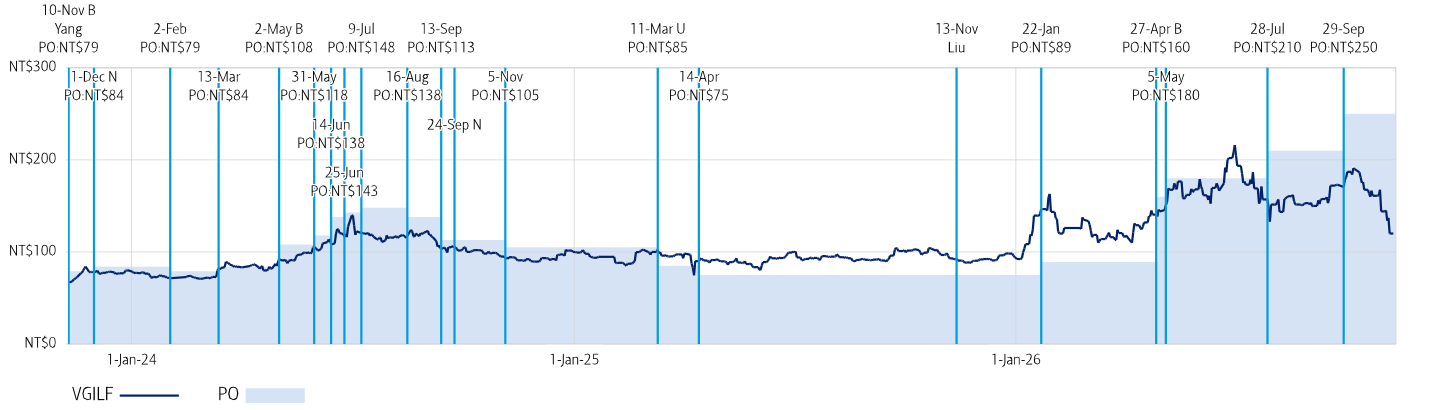
iQdatabase® is our real-time global research database that is sourced directly from our equity analysts' earnings models and includes forecasted as well as historical data for income statements, balance sheets, and cash flow statements for companies covered by BofA Global Research.

iQprofile℠, *iQmethod℠* are service marks of Bank of America Corporation. *iQdatabase®* is a registered service mark of Bank of America Corporation.

Disclosures

Important Disclosures

Vanguard (VIS) (VGILF) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

FUNDAMENTAL EQUITY OPINION KEY: Opinions include a Volatility Risk Rating, an Investment Rating and an Income Rating. VOLATILITY RISK RATINGS, indicators of potential price fluctuation, are: A - Low, B - Medium and C - High. INVESTMENT RATINGS reflect the analyst's assessment of both a stock's absolute total return potential as well as its attractiveness for investment relative to other stocks within its Coverage Cluster (defined below). Our investment ratings are: 1 - Buy stocks are expected to have a total return of at least 10% and are the most attractive stocks in the coverage cluster; 2 - Neutral stocks are expected to remain flat or increase in value and are less attractive than Buy rated stocks and 3 - Underperform stocks are the least attractive stocks in a coverage cluster. An investment rating of 6 (No Rating) indicates that a stock is no longer trading on the basis of fundamentals. Analysts assign investment ratings considering, among other things, the 0-12 month total return expectation for a stock and the firm's guidelines for ratings dispersions (shown in the table below). The current price objective for a stock should be referenced to better understand the total return expectation at any given time. The price objective reflects the analyst's view of the potential price appreciation (depreciation).

Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2] Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

INCOME RATINGS, indicators of potential cash dividends, are: 7 - same/higher (dividend considered to be secure), 8 - same/lower (dividend not considered to be secure) and 9 - pays no cash dividend. *Coverage Cluster* is comprised of stocks covered by a single analyst or two or more analysts sharing a common industry, sector, region or other classification(s). A stock's coverage cluster is included in the most recent BofA Global Research report referencing the stock.

Price Charts for the securities referenced in this research report are available on the Price Charts website, or call 1-800-MERRILL to have them mailed.
In the US, retail sales and/or distribution of this report may be made only in states where these securities are exempt from registration or have been qualified for sale: Vanguard (VIS).
The country in which this issuer is organized has certain laws or regulations that limit or restrict ownership of the issuer's shares by nationals of other countries: Vanguard (VIS).
BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel

expenses from the issuer for such visits.
Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.
The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.
Officers of BofAS or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.
Refer to BofA Global Research policies relating to conflicts of interest.
"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.
Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:
BofAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at www.bofaml.com/BofASEdisclaimer; BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSFC); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority.
This information: has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSFC; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the Electronic Communications Disclaimers for further information.
This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.
General Investment Related Disclosures:
Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.
This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives, financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.
Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are not regulated in a manner similar to U.S. securities markets. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.
This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may

be more or less positive than a stock's fundamental equity rating.
BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.
Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.
BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities or digital assets that may have been recommended or discussed in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments recommended or discussed in this report.
BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.
In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQdatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.
Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.
This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.
Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.
Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.
The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.
All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.
Subject to the quiet period applicable under laws of the various jurisdictions in which we distribute research reports and other legal and BofA Securities policy-related restrictions on the publication of research reports, fundamental equity reports are produced on a regular basis as necessary to keep the investment recommendation current.
Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.
In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.
Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.

